

2026.9.7

以下 $\subset$ 和 $\subset$ 均包含 " $=$ " 的情形

Algebra is about Structures.

eg. Field $(F, +, \cdot)$ + operation laws/axioms

Vector space $(V, +, \cdot)$ + axioms

↓ 同态
Endomorphism algebra $(\text{End } V, +, \cdot, \circ)$

§1. Groups, isomorphism, subgroups

DEF A group is a pair $(G, \cdot)$ where G : nonempty set $\cdot: G \times G \rightarrow G$ binary operation

st. 1) (associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

2) $\exists$ an element $e \in G$, called an identity of G , st. $a \cdot e = e \cdot a = a, \forall a \in G$.

3) $\forall a \in G, \exists$ an element $a^{-1} \in G$, called an inverse of a , st. $a \cdot a^{-1} = a^{-1} \cdot a = e$

The group G is called abelian or commutative if $a \cdot b = b \cdot a \forall a, b \in G$.

REM without 3) $\rightarrow$ get the definition of a monoid. 么半群

without 2), 3) $\rightarrow$ a semigroup.

eg. $(\mathbb{Z}, +)$ identity 0.

$(\mathbb{Q} \setminus \{0\}, \cdot)$ identity 1.

$(\mathbb{Z}/n\mathbb{Z}, +)$

$$a * b + 1 = (a+1)(b+1) \quad \phi(a * b) = \phi(a) \phi(b)$$

$(\mathbb{Q} \setminus \{-1\}, *)$ $a * b = ab + a + b$ in fact, shifted from $(\mathbb{Q} \setminus \{0\}, \cdot)$ by 1 hence the "same".

$(G, \cdot)$ and $(H, \circ)$ both groups.

define $(G \times H, *)$ $(a, a') * (b, b') = (a \cdot b, a' \circ b')$ called the direct product of $(G, \cdot)$ and $(H, \circ)$.

Basic properties of groups

1) the identity is unique. $(e = e \cdot e' = e')$

2) the inverse of a is unique. $(a^{-1} = a^{-1} \cdot a \cdot (a^{-1})' = (a^{-1})')$

3) $(a^{-1})^{-1} = a$

4) $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$

5) $a \cdot u = a \cdot v \Rightarrow u = v$ $u \cdot b = v \cdot b \Rightarrow u = v$

Notation/conclusion

1) Omit $\cdot$ when safe

eg. say "let G be a group" ab (instead of $a \cdot b$) $a^n = \overbrace{a \cdots a}^{n \text{ times}}$

2) View $\cdot$ as "multiplication" and often use 1 for the identity

3) Additive convention: when we know G is abelian

use $+$ for the group operation

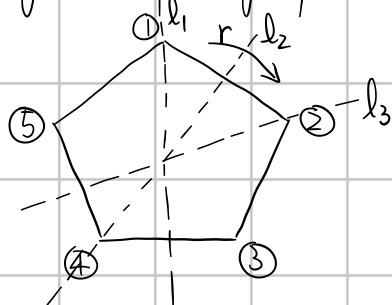
0 for the identity

$-a$ for the inverse of a

eg. $a+b=b+a$ $na = \overbrace{a+\cdots+a}^{n \text{ times}}$

Further examples

eg. (Dihedral groups) 二面体群



$D_{2n} := \{ \text{symmetries of a regular } n\text{-gon} \}$
有时记号是 D_n
 nonabelian for $n \geq 3$

elements $e = \text{identity}$ $r_1 = \text{rotation } \frac{2\pi}{n}$ $r_2 = \text{rotation } \frac{4\pi}{n}$...
↕
 $s_1 = \text{reflection about } l_1$ $s_2 = \text{reflection about } l_2$...

$\#D_{2n} = 2n$

More efficiently,

group operation = composition

$$D_{2n} = \{ 1, r=r_1, r^2, \dots, r^{n-1} \} \cup \{ s=s_1, rs, r^2s, \dots, r^{n-1}s \}$$

first reflect about l_1 , then rotate $\frac{2\pi}{n}$

想象时可追踪顶点

$$= \langle r, s \mid r^n = 1, s^2 = 1, srs = r^{-1} \rangle$$

generators

relations

群的"字母表"

DEF A subset $S = \{s_1, \dots, s_n\}$ of G is called a set of generators

if every element of G can be written as a finite product of $s_1, \dots, s_n, s_1^{-1}, \dots, s_n^{-1}$ called a "word"

An equality consisting of generators and their inverses is called a "relation".

We write $G = \langle s_1, \dots, s_n \mid R_1, \dots, R_m \rangle$ if all relations in G can be derived from $R_1, \dots, R_m$.

↓
called a presentation of G .

eg. $\mathbb{Z}/6\mathbb{Z} = \langle s \mid s^6 = 1 \rangle$

$$= \langle r, s \mid r^3 = 1, s^2 = 1, rs = sr \rangle$$

eg. (Symmetric groups/permutation groups)

DEF Let Ω be a set. Then $S_\Omega = \{ \text{bijections } \sigma: \Omega \rightarrow \Omega \}$ has the structure of a group,

called the symmetric group/permutation on the group on the set Ω .

for $\Omega = \{1, 2, \dots, n\}$ write S_n . $\#S_n = n!$ nonabelian for $n \geq 3$.

Elements of S_n .

expression 1: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 5 & 1 & 3 & 2 & 6 & 4 \end{pmatrix}$

expression 2: $\begin{array}{c} \overbrace{1 \mapsto 7 \mapsto 4 \mapsto 3} \\ \underbrace{2 \mapsto 5} \\ \downarrow 6 \end{array}$ write $\sigma = (1\ 7\ 4\ 3)(2\ 5)(6)$

here $(a_1 a_2 \dots a_r)$ is called a cycle which means $a_1 \mapsto a_2 \mapsto \dots \mapsto a_r$ with all other a_i 's untouched.

↓
all distinct

Easy facts.

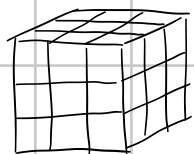
- every element of S_n can be written as a product of disjoint cycles.
- disjoint cycles commute with each other.

Exercise S_n is generated by $\begin{cases} \text{all transpositions } (i, j) \\ \text{all adjacent transpositions } (i, i+1) \end{cases}$ → 类似冒泡排序
 $(1, 2), (1, 2, \dots, n)$ → 通过这2个就可生成

REM $\forall n \geq 3$, D_n can be viewed as a "subgroup" (we'll define) of S_n .

eg. (Rubik's cube group)

6个中心块固定 $6 \times (9-1)$



can be viewed as a "subgroup" of S_{48} generated by 6 elements FBUDLR.

$\# = 43252003274489856000$.

God's number (2014): every element can be written as a word of length ≤ 26 . → 即所有魔方状态可用至多26步还原

Group isomorphisms

DEF Two groups $(G, \cdot)$ and $(H, \circ)$ are isomorphism, if $\exists$ bijection $\Phi: G \xrightarrow{\sim} H$, st.

$$\forall a, b \in G,$$

$$1) \Phi(a \cdot b) = \Phi(a) \circ \Phi(b)$$

同构 $\rightarrow$ 乘法表相同, 只是重新命名了元素.

$$\left(\begin{array}{l} 2) \Phi(e_G) = e_H \\ 3) \Phi(a^{-1}) = \Phi(a)^{-1} \end{array} \right) \quad 1) \Rightarrow 2) \Rightarrow 3)$$

$$\text{Write } G \cong H \quad \text{or} \quad G \xrightarrow[\sim]{\Phi} H$$

eg. $(\mathbb{R}, +) \xrightarrow[\sim]{\exp} (\mathbb{R}_{>0}, \cdot)$

$$(\mathbb{Z}/n\mathbb{Z}, +) \xrightarrow{\sim} (\mu_n = \{n^{\text{th}} \text{ roots of unity in } \mathbb{C}\}, \cdot)$$

$$a \mapsto e^{\frac{2\pi i a}{n}}$$

REM Fundamental question in group theory: classify all groups up to isomorphisms.

eg. all groups of order 6 are isomorphic to either $\mathbb{Z}/6\mathbb{Z}$ or S_3 .

$$\left(\begin{array}{l} \Rightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/6\mathbb{Z} \\ D_6 \cong S_3 \end{array} \right) \rightarrow \text{nonabelian}$$

Subgroups

DEF A subset H of a group G is a subgroup denoted $H < G$.

$$\text{iff } \begin{array}{l} 1) e \in H \\ 2) \forall a, b \in H, ab \in H \\ 3) \forall a \in H, a^{-1} \in H \end{array}$$

Equivalently, a nonempty subset $H \subset G$ is a subgroup $\Leftrightarrow \forall a, b \in H, ab^{-1} \in H$.

DEF Let G be a group and $A \subset G$ a subset. $\langle A \rangle :=$ subgroup of G generated by A .

$$\underbrace{= \{a_1^{\pm 1} \dots a_r^{\pm 1} \mid a_1, \dots, a_r \in A\}}_{\text{finite product}} = \bigcap_{\substack{\text{subgroup } H \subset G \\ \text{st. } A \subset H}} H$$

DEF A group H is called cyclic if it can be generated by one element.

ie. $\exists x \in H$, st. $H = \langle x \rangle = \{x^n \mid n \in \mathbb{Z}\}$

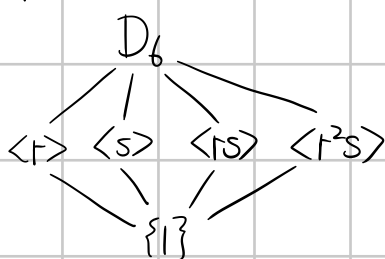
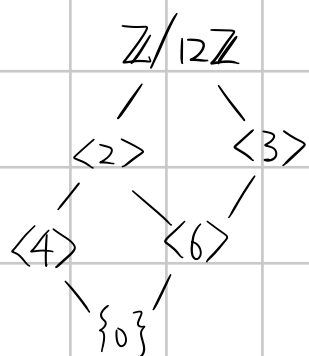
Two cases.

1) $\#H = n$ then $H = \{1, x, x^2, \dots, x^{n-1}\}$ cyclic of order n . $H \cong \mathbb{Z}/n\mathbb{Z}$
 $x^a \mapsto a$.

2) $\#H = \infty$ then $H = \{1, x, x^2, \dots, x^{-1}, x^{-2}, \dots\} \cong \mathbb{Z}$.

DEF Let G be a group and $x \in G$. Define the order of x in G . $|x| := \#\langle x \rangle \in \mathbb{Z}_{>0} \cup \{\infty\}$.

Picture: lattices of subgroups.



Analogy

Vector spaces

Groups

(finite) direct sum

(finite) direct product

isomorphism

isomorphism

subspace

把一个线性子空间整体平移

subgroup

affine subspace

coset

quotient space

quotient group

linear map

homomorphism

eg. (quaternion group) $Q = \{\pm 1, \pm i, \pm j, \pm k\}$

$$\boxed{i^2 = j^2 = k^2 = ijk = -1}$$

